

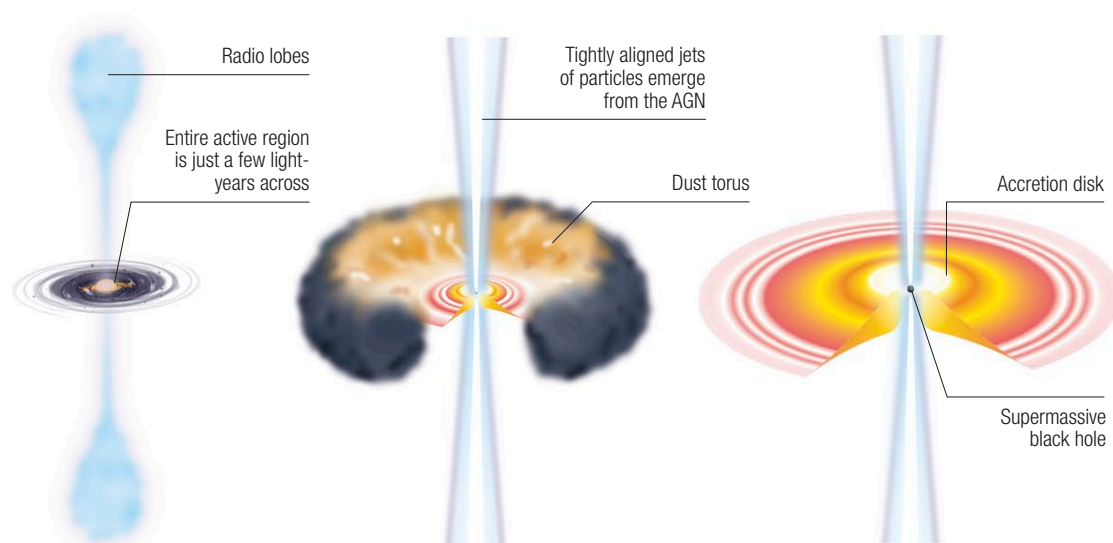
ACTIVE GALAXIES

MANY GALAXIES SHOW SIGNS OF ENERGY OUTPUT FROM THEIR CENTRAL REGIONS THAT CANNOT BE EXPLAINED BY STARS ALONE. THESE ACTIVE GALAXIES APPEAR VARIED, BUT ALL CAN BE EXPLAINED BY THE SAME MECHANISM.

Active galaxies generate vast amounts of energy—not only visible light but also as radio waves, X-rays, ultraviolet radiation, and gamma rays—from regions near their center. These galaxies are divided into four major types: Seyfert galaxies, radio galaxies, quasars, and blazars. Seyfert galaxies are otherwise normal-looking spiral galaxies with a bright and concentrated source of radiation embedded in their center, and unusual overall energy output that cannot be explained through starlight alone. Radio galaxies, in contrast, are distinguished by two large clouds of radio-emitting gas on either side of a central galaxy (in some cases, narrow jets can be seen linking them to the heart of the galaxy). Quasars are very distant galaxies with an intense starlike source of light, far brighter than a Seyfert galaxy, at their center. They are also sources of radio waves,

and vary in brightness over hours or days. Finally, blazars are broadly similar to quasars, but have distinctive differences in their radiation that mark them out.

Astronomers think that all these different types of activity are actually caused by the same kind of object—an engine called an active galactic nucleus (AGN). The speed at which AGNs change their energy output means that they can be no larger than our Solar System, and the only object capable of releasing such vast amounts of energy in such a small region of space is a superheated “accretion disk” of matter falling into a supermassive black hole. The amount of material falling into the black hole, and the AGN’s alignment as seen from our point of view on Earth, determine which type of active galaxy we see.



△ Whole galaxy

From a distance, an active galaxy may be surrounded by two huge lobes of radio emission, created as jets of particles ejected from the central disk encounter gas in intergalactic space. When the central AGN is visible, its light output can dwarf that of the surrounding galaxy.

△ Dust torus

The active galaxy’s central regions are surrounded by a thick doughnut-shaped ring or torus of light-obscuring gas and dust. If we see this ring edge-on, then the surrounding radio lobes are the only visible sign of unusual activity, manifesting as a radio galaxy.

△ Nucleus

At the heart of the active galactic nucleus is a supermassive black hole. With the gas of a million or more Suns, it pulls material to its doom, heating it to millions of degrees where it emits intense radiation. If the particle jet points directly toward Earth, the AGN creates a blazar.

